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Remifentanil in the postoperative pain treatment after Nuss procedures – our experience

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Introduction: Minimally invasive surgical techniques have increasingly been used to repair pectus deformities. Perioperative anesthetic and pain management of patients undergoing the Nuss procedure is a challenge to the anesthesiologist. A general anesthesia supplemented with adequate analgesic technique is crucial to the success of the surgery and reducing perioperative complications.

Material and methods: On the Clinic of pediatric surgery in Novi Sad, Nuss procedure was conducted in 38 patients, average age 15,2 years, in period from 2006-2019. All the patients had a uniform premedication with intramuscular midazolam and atropine. Anesthesia was performed using propofol/sevofluran, analgesia (intra and postoperative) was achieved with continuous infusion of remifentanyl or with epidural anesthesia, and we used boluses of muscle relaxing agent. Pain assessment was performed using numerical pain scale.

Results: Anesthesia was maintained with general anesthesia using either remifentanyl + propofol - in 25 (65,8%) patients, or sevoflurane + remifentanyl – in 13 (34,2%) patients. Epidural anesthesia was performed in 9 patients (23,7%). Postoperative analgesia was achieved with epidural (chirocaine + fentanyl) - 9 patients (23,7%) for 48h, or continuous remifentanyl (0,05mcg / kg / min, for 24h-36h) in 29 patients (76,3%). All patients, after that period, received tramadol + NSAID. Pain score values were 2 – 4 in all patients. Average hospitalization for all patients was 8,6 days, with no difference regarding to type of analgesia.

Conclusion: NUSS operation is a very painful procedure performed quite frequently in adolescents. Continuous intraoperative and postoperative remifentanyl infusion provide good analgesia and early mobilization, so therefore seems as a good and less invasive alternative to epidural analgesia for these procedures.

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